

| Course Type | Course Code | Course Name                 | Credits |
|-------------|-------------|-----------------------------|---------|
| PEC         | CEPEC7032   | NATURAL LANGUAGE PROCESSING | 03      |

| Examination Scheme     |                         |                         |                      |     |             |
|------------------------|-------------------------|-------------------------|----------------------|-----|-------------|
| Distribution of Marks  |                         |                         | Exam Duration (Hrs.) |     | Total Marks |
| In-semester Assessment |                         | End Semester Exam (ESE) |                      |     |             |
| Continuous Assessment  | Mid-Semester Exam (MSE) |                         | MSE                  | ESE |             |
| 20                     | 30                      | 50                      | 1.5                  | 2   | 100         |

**Pre-requisite:**

1. CEPCC303-Data Structures
2. CEPCC509 -Theory of Computer Science
3. CEPCC405- Engineering Mathematics-IV

**Program Outcomes addressed:**

1. PO1: Engineering knowledge.
2. PO2: Problem analysis.
3. PO3: Design/development of solutions.
4. PO4: Conduct investigations of complex problems
5. PO5: Engineering Tools usage
6. PO8: Individual and Collaborative Team Work

**Course Objectives:**

1. To inculcate learners to the basic concepts, linguistic foundations, key processing stages, and real-world relevance of Natural Language Processing.
2. To familiarize text preprocessing, morphological analysis, and statistical word-level modeling in NLP.
3. To guide learners to leverage syntactic structures and sequence modeling using POS tagging, grammar, and neural models.
4. To facilitate the learners to apply semantic representation, word sense disambiguation, and embedding models for meaning extraction in NLP.
5. To instill discourse structure, reference resolution, and pragmatic aspects of language, and to explore evaluation techniques for NLP systems.
6. To equip the learners to apply NLP techniques to design real world NLP applications.

| Module     | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Hrs          |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|            | <b>Course Introduction</b><br>This course introduces the fundamental concepts and modern techniques of Natural Language Processing (NLP). It focuses on facilitating the learners to enable machines to process and understand human language for applications such as translation, summarization, and sentiment analysis, fostering skills essential for AI and data-driven language technologies.                                                                                                                                                                                                                                                           | <b>01</b>    |
| <b>01.</b> | <b>Introduction to NLP</b><br><b>Learning Objective/s:</b><br><i>Learners are expected to explore the fundamental concepts of Natural Language Processing, focusing on its linguistic foundations, stages of processing, types of ambiguities, major challenges, and real-world applications.</i>                                                                                                                                                                                                                                                                                                                                                             | <b>2-4</b>   |
|            | <b>Contents:</b><br>1.1 NLP Pipeline and Levels of Analysis: Stages in NLP processing-text preprocessing, morphological, syntactic, semantic, pragmatic, and application-level stages.<br>1.2 Ambiguity and Challenges in NLP (English and Indian Languages)<br>1.3 Applications of NLP<br>1.4 Tools and Resources: NLTK, SpaCy, Hugging Face                                                                                                                                                                                                                                                                                                                 |              |
|            | <b>Self-Learning Topics:</b><br><i>NLP for Indian Languages, Unicode and Tokenizers, variety of tools for regional language preprocessing and data formats</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              |
|            | <b>Learning Outcomes:</b><br><i>A learner will be able to</i><br><br><i>LO 1.1: Apply fundamental mathematical and statistical concepts to understand the NLP pipeline. (P.I. – 1.1.1)</i><br><i>LO 1.2: Apply core computer science principles to the different stages of NLP processing. (P.I. – 1.4.1)</i><br><i>LO 1.3: Identify and describe the key modules of an NLP system, including preprocessing, morphological, syntactic, semantic, pragmatic, and application-level analysis. (P.I. – 2.1.2)</i><br><i>LO 1.4: Analyze the limitations and challenges of NLP solutions, including ambiguity in English and Indian languages. (P.I. – 2.4.3)</i> |              |
| <b>02.</b> | <b>Text Processing and Word-Level Analysis</b><br><b>Learning Objective/s:</b><br><i>To study the processes involved in text preprocessing, lexical and morphological analysis, and to understand the use of regular expressions, finite-state methods, and statistical models such as N-grams for word-level processing and representation in Natural Language Processing.</i>                                                                                                                                                                                                                                                                               | <b>10-12</b> |

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|     | <p><b>Contents:</b></p> <p>2.1 Lexical Analysis: Basics of lexical processing and word representation.</p> <p>2.2 Empirical Laws on Raw Text</p> <p>2.3 Preprocessing tasks: Sentence segmentation, tokenization, normalization, stop-word removal, stemming, lemmatization</p> <p>2.4 Morphological Analysis: Inflectional and derivational morphology</p> <p>2.5 Regular Relations and Finite-State Morphology: Regular expressions, finite-state automata, and transducers for morphological analysis.</p> <p>2.6 Stemming and Lemmatization: Porter and Snowball algorithms; dictionary- and rule-based lemmatization.</p> <p>2.7 Statistical Models</p> <p>2.7.1 N-gram Models: unigram, bigram, trigram, and their evaluation.</p> <p>2.7.2 Smoothing Techniques: Laplace, Good–Turing</p> <p>2.7.3 Evaluation Metrics: Perplexity</p> <p>2.8 Word Representation Models: Bag of Words and TF-IDF features for statistical text representation.</p> <p><b>Self-Learning Topics:</b><br/> Edit Distance and Noisy Channel Models for text normalization and spelling correction</p> <p><b>Learning Outcomes:</b><br/> A learner will be able to</p> <p>LO 2.1: Apply mathematical models to analyze lexical processing and word-level patterns. (P.I. – 1.1.1)</p> <p>LO 2.2: Apply core computer science principles to implement text analysis tasks such as tokenization, stemming, lemmatization, and morphological processing. (P.I. – 1.4.1)</p> <p>LO 2.3: Recognize and apply relevant mathematical and algorithmic techniques to analyze raw text and interpret linguistic patterns.(P.I. – 2.1.3)</p> <p>LO 2.4: Implement NLP algorithms using engineering mathematics, including finite-state automata and regular relations for morphological analysis. (P.I. – 2.4.1)</p> <p>LO 2.5: Define a clear and precise problem statement for lexical, preprocessing, or morphological text processing tasks, including appropriate objectives and scope. (P.I. – 3.1.1)</p> <p>LO 2.6: Design, compare, and evaluate multiple text preprocessing and feature extraction pipelines, including Bag-of-Words and TF-IDF representations. (P.I. – 3.2.2)</p> |      |
| 03. | <p><b>Syntax Analysis and Sequence Modeling</b></p> <p><b>Learning Objective/s:</b><br/> Expected to analyse and apply parallel database architectures and I/O parallelism techniques to design and optimize efficient database systems.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 9-11 |

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|     | <p><b>Contents:</b></p> <p>3.1 Part-of-Speech (POS) Tagging: Stochastic based approach; POS tagging using Hidden Markov Model (HMM) and Viterbi algorithm; challenges in POS tagging.</p> <p>3.2 Discriminative Models for Sequence Labeling: Maximum Entropy Markov Model (MEMM) and Conditional Random Field (CRF), BiLSTM-CRF for neural sequence labeling.</p> <p>3.3 Grammar Formalisms: Context-Free Grammar (CFG) and Probabilistic Context-Free Grammar (PCFG); modelling syntactic structure and constituency.</p> <p>3.4 Parsing Techniques:</p> <p>    3.4.1 Top-Down Parsers: Predictive and Early Parser;</p> <p>    3.4.2 Bottom-Up Parsers: Shift-Reduce and CKY algorithm;</p> <p>    3.4.3 Introduction to Dependency and Constituency Parsing.</p> <p>3.5 Neural Sequence Models:</p> <p>Introduction to Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU) models for syntactic sequence processing; overview of Attention Mechanisms and Transformers for modern NLP parsing tasks.</p> <hr/> <p><b>Self-Learning Topics:</b></p> <p><i>Transformer-based models (BERT, GPT) for sequence labeling and syntactic parsing.</i></p> <hr/> <p><b>Learning Outcomes:</b></p> <p><i>Learner should be able to</i></p> <p><i>LO 3.1: Apply mathematical and statistical concepts to model syntactic structures and sequence-labeling methods in NLP. (P.I. – 1.1.1)</i></p> <p><i>LO 3.2: Apply core computer science principles to model syntactic structures and sequence labeling tasks. (P.I. – 1.4.1)</i></p> <p><i>LO 3.3: Implement syntactic and sequence modeling solutions using mathematical and computational models such as HMMs, CRFs, and neural architectures. (P.I. – 2.4.1)</i></p> <p><i>LO 3.4: Identify the limitations and sources of errors in syntactic and sequence labeling solutions. (P.I. – 2.4.3)</i></p> <p><i>LO 3.5: Define a clear problem statement with appropriate objectives and scope for syntax-related NLP tasks. (P.I. – 3.1.1)</i></p> <p><i>LO 3.6: Design and evaluate alternative approaches for POS tagging, parsing, and sequence labeling tasks. (P.I. – 3.2.2)</i></p> |     |
| 04. | <p><b>Semantic Analysis and Word Representations</b></p> <p><b>Learning Objective/s:</b></p> <p><i>To understand semantic representation and lexical relations, apply word sense disambiguation techniques, and explore vector-based and contextual embedding models for capturing word meaning in NLP.</i></p> <hr/> <p><b>Contents:</b></p> <p>4.1 Semantic Analysis: Meaning representation, lexical semantics, and relations among words: synonymy, polysemy, hyponymy, meronymy, and semantic ambiguity.</p> <p>4.2 Word Sense Disambiguation (WSD): Approaches to WSD: knowledge-based (Lesk's Algorithm), supervised (Naïve Bayes), and unsupervised (clustering-based).</p> <p>(Classical Approaches, Intermediate / Machine Learning-Enhanced Approaches, Deep Learning-Based / Contextual WSD)</p> <p>4.3 Vector Semantics and Word Embeddings: Vectorization techniques for representing meaning; distributional semantics and word embeddings: Word2Vec and GloVe; introduction to subword embeddings (FastText) and contextual embeddings (BERT).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 7-9 |

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|     | <p><b>Self-Learning Topics:</b><br/>Transformer-based models (BERT, RoBERTa, GPT) for contextual and multilingual semantic representation.</p> <hr/> <p><b>Learning Outcomes:</b><br/>A learner will be able to<br/> LO 4.1: Apply mathematical and statistical concepts to analyze semantic meaning, word relations, and word-sense patterns in NLP. (P.I. – 1.1.1)<br/> LO 4.2: Apply computer science and AI principles to understand semantic analysis and meaning representation. (P.I. – 1.4.1)<br/> LO 4.3: Implement semantic and word-sense models using appropriate mathematical formulations. (P.I. – 2.4.1)<br/> LO 4.4: Analyze the limitations of semantic models used in NLP. (P.I. – 2.4.3)<br/> LO 4.5: Able to choose appropriate procedures, algorithms, and test cases for semantic analysis tasks. (P.I. – 4.1.2)<br/> LO 4.6: Interpret data-driven trends in Word Sense Disambiguation and word embedding techniques. (P.I. – 4.3.2)<br/> LO 4.7: Create tools or workflows for word embeddings and semantic modeling. (P.I. – 5.1.2)<br/> LO 4.8: Adapt modern NLP frameworks to implement and refine semantic and embedding-based models. (P.I. – 5.2.2)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |
| 05. | <p><b>Discourse and Advanced NLP</b></p> <p><b>Learning Objective/s:</b><br/>To understand discourse structure, reference and coreference resolution, and pragmatic aspects of language, and to explore modern neural approaches and evaluation metrics in NLP.</p> <hr/> <p><b>Contents:</b></p> <p>5.1 Discourse Structure and Coherence:<br/>Discourse relations, cohesion and coherence in connected text; syntactic and semantic constraints on discourse interpretation.</p> <p>5.2 Reference and Coreference Resolution:<br/>Reference phenomena and resolution techniques; anaphora resolution using Hobbs' Algorithm and Centering Theory; introduction to neural coreference models in NLP.</p> <p>5.3 Pragmatics in NLP:<br/>Pragmatic aspects of language :implicature, presupposition, and speech acts in computational interpretation.</p> <p>5.4 Evaluation Metrics in NLP:<br/>Evaluation of NLP systems using BLEU, ROUGE, and F1-score metrics.</p> <hr/> <p><b>Self-Learning Topics:</b><br/>Transformer-based and neural models for coreference resolution and discourse understanding.</p> <hr/> <p><b>Learning Outcomes:</b><br/>A learner will be able to</p> <p>LO 5.1: Apply mathematical techniques to analyze and evaluate discourse and coreference resolution methods. (P.I. – 1.1.1)<br/> LO 5.2: Apply theoretical principles of computer science and linguistic knowledge to analyze discourse structure and coherence. (P.I. – 1.4.1)<br/> LO 5.3 Apply algorithmic and engineering mathematics principles to implement discourse and coreference resolution techniques. (P.I. – 2.4.1)<br/> LO 5.4: Analyze evaluation metrics to assess NLP system performance. (P.I. – 2.4.3)<br/> LO 5.5: Choose appropriate algorithms and procedures for reference and coreference resolution tasks. (P.I. – 4.1.2)<br/> LO 5.6: Identify strengths, limitations, and trends revealed through metric-based evaluation. (P.I. – 4.3.2)</p> | 5-7 |

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| 06.          | <p><b>Applications and Ethical Perspectives in NLP</b></p> <p><i><b>Learning Objective/s:</b></i><br/>To study key NLP applications such as machine translation, summarization, sentiment analysis, and question answering, and to understand ethical considerations and challenges in low-resource language processing.</p> <p><b>Contents:</b><br/>         6.1 Machine Translation: Rule-based, statistical, and neural approaches; introduction to transformer-based models.<br/>         6.2 Text Summarization: Extractive vs. abstractive summarization; evaluation metrics.<br/>         6.3 Sentiment Analysis: Lexicon-based and supervised learning approaches for opinion mining.<br/>         6.4 Information Extraction and Question Answering: Entity recognition, relation extraction, and retrieval-augmented generation (RAG) for QA systems.<br/>         6.5 Ethics and Societal Impact in NLP: Bias and fairness in models, privacy concerns, hallucinations, misinformation, and cultural inclusivity.<br/>         6.6 NLP for Low-Resource and Regional Languages: Challenges and tools for Indian and underrepresented languages.</p> <p><i><b>Self-Learning Topics:</b></i><br/>Conversational AI and multilingual NLP applications using transformer-based models.</p> <p><i><b>Learning Outcomes:</b></i><br/>         A learner will be able to<br/>         LO 6.1: Apply mathematical and statistical foundations to understand and analyze core components of complex NLP systems.(P.I. 1.1.1)<br/>         LO 6.2: Apply principles of computer science and engineering to design and evaluate solutions for diverse NLP tasks.(P.I. 1.4.1)<br/>         LO 6.3: Identify the key components and processing steps involved in advanced NLP applications to determine requirements for solving a given task.(P.I. -2.1.2)<br/>         LO 6.4: Identify the limitations and potential sources of error in advanced NLP applications (P.I. -2.4.3)<br/>         LO 6.5: Define a clear problem statement with appropriate objectives and scope for advanced NLP tasks. (P.I. -3.1.1)<br/>         LO 6.6: Generate multiple design approaches to address functional requirements in NLP applications.(P.I. -3.2.2)<br/>         LO 6.7: Create or adapt tools and techniques to address NLP problems. (P.I. -5.1.2)<br/>         LO 6.8: Elaborate effective use of NLP tools and frameworks. (P.I. -5.2.2)<br/>         LO 6.9: Demonstrate effective communication, collaborative problem-solving, and teamwork in NLP tasks.(P.I. -8.2.1)<br/>         LO 6.10: Present integrated team results with clear coordination and contribution from all members.(P.I. -8.3.1)</p> | 4-6       |
|              | <p><b>Course Conclusion</b></p> <p>This course offers an integrated understanding of linguistic theory and computational methods in Natural Language Processing, enabling students to apply classical and neural approaches to analyze and model language, and to develop responsible NLP solutions for real-world applications. It builds a strong foundation for advanced research and innovation in artificial intelligence and human language technologies.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 01        |
| <b>Total</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>45</b> |

**Performance Indicators:**

| <b><u>P.I. No.</u></b> | <b><u>P.I. Statement</u></b>                                                                                       |
|------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1.1.1                  | Apply the knowledge of discrete structures, linear algebra, statistics and numerical techniques to solve problems. |
| 1.4.1                  | Apply theory and principles of computer science and engineering to solve an engineering problem.                   |
| 2.1.2                  | Identify processes/modules/algorithms of a computer-based system and parameters to solve a problem.                |
| 2.1.3                  | Identify mathematical algorithmic knowledge that applies to a given problem.                                       |
| 2.4.1                  | Applies engineering mathematics to implement the solution.                                                         |
| 2.4.3                  | Identify the limitations of the solution and sources/ causes.                                                      |
| 3.1.1                  | Able to define a precise problem statement with objectives and scope.                                              |
| 3.2.2                  | Able to produce a variety of potential design solutions suited to meet functional requirements.                    |
| 4.1.2                  | Able to choose appropriate procedure/algorithm, dataset and test cases.                                            |
| 4.2.1                  | Design and develop appropriate procedures/methodologies based on the study objectives.                             |
| 4.3.2                  | Critically analyze data for trends, stating possible errors and limitations.                                       |
| 5.1.2                  | Create/adapt/modify/extend tools and techniques to solve engineering problems.                                     |
| 5.2.2                  | Demonstrate proficiency in using discipline-specific tools.                                                        |
| 8.2.1                  | Demonstrate effective communication, problem-solving, teamwork.                                                    |
| 8.3.1                  | Present results as a team, with smooth integration of contributions from all individual efforts.                   |

**Course Outcomes:** Learner will be able to-

1. Develop a basic understanding of Natural Language Processing with regards to knowledge, ambiguity, stages, challenges and applications in NLP for effective analysis of the NLP pipeline.(LO 1.1, LO 1.2, LO 1.3, LO 1.4)
2. Apply text preprocessing techniques, perform morphological analysis, and use statistical models for word-level language processing. (LO 2.1, LO 2.2, LO 2.3, LO 2.4, LO 2.5, LO 2.6)
3. Design and evaluate computational approaches for syntactic and sequence-labeling tasks in NLP. (LO 3.1,LO 3.2, LO 3.3, LO 3.4, LO 3.5, LO 3.6)
4. Develop strategies to model and analyze meaning in language using semantic relations, word-sense techniques, and embedding-based representations.(LO 4.1, LO 4.2, LO4.3, LO 4.4, LO 4.5, LO 4.6, LO 4.7, LO 4.8)
5. Apply discourse and pragmatic analysis using reference resolution techniques and evaluate NLP systems using standard metrics.(LO 5.1, LO 5.2, LO 5.3, LO 5.4, LO 5.5, LO 5.6)
6. Develop solutions for various NLP problems.(LO 6.1, LO 6.2, LO 6.3, LO 6.4, LO 6.5, LO 6.6, LO 6.7, LO 6.8, LO 6.9, LO 6.10)

**CO-PO Mapping Table with Correlation Level**

| CO ID          | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CEPE5012.1     | 3   | 3   | --  | --  | --  | --  | --  | --  | --  | --   | --   |
| CEPE5012.2     | 3   | 3   | 3   | --  | --  | --  | --  | --  | --  | --   | --   |
| CEPE5012.3     | 3   | 3   | 3   | --  | --  | --  | --  | --  | --  | --   | --   |
| CEPE5012.4     | 3   | 3   | --  | 3   | 3   | --  | --  | --  | --  | --   | --   |
| CEPE5012.5     | 3   | 3   | --  | 3   | --  | --  | --  | --  | --  | --   | --   |
| CEPE5012.6     | 3   | 3   | 3   | --  | 3   | --  | --  | 3   | --  | --   | --   |
| <b>Average</b> | 3   | 3   | 3   | 3   | 3   | --  | --  | 3   | --  | --   | --   |

**Text Books:**

1. Daniel Jurafsky, James H. and Martin, Speech and Language Processing, Second Edition, Prentice Hall, 2008.
2. Christopher D. Manning and Hinrich Schütze, Foundations of Statistical Natural Language Processing, MIT Press, 1999.

**Reference Books:**

1. Alexander Clark, Chris Fox, and Shalom Lappin, The Handbook of Computational Linguistics and Natural Language Processing, Wiley-Blackwell, 2010
2. Steven Bird, Ewan Klein, and Edward Loper, Natural Language Processing with Python (NLTK Book), O'Reilly Media, 2nd Edition, 2021
3. Jacob Eisenstein, Introduction to Natural Language Processing, MIT Press, 2019
4. Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, and Harshit Surana, Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems, O'Reilly Media, 2020

**Other Resources:**

1. <https://www.coursera.org/specializations/natural-language-processing>
2. <http://www.cse.iitb.ac.in/~cs626-449>
3. <http://cse24-iiith.virtual-labs.ac.in/#>
4. <https://nptel.ac.in/courses/106105158>

**IN-SEMESTER ASSESSMENT (50 MARKS)****1. Continuous Assessment - Theory-(20 Marks)*****Suggested breakup of distribution***

Assignment on Live problems / Case Studies wherein problem statements such as text classification, sentiment analysis, or part-of-speech tagging are given prior. Students are expected to research and collect relevant linguistic datasets and resources. They can use these resources and solve the assigned NLP problem on the specified date and time in Institute premises in the presence of a faculty member.:

10 Marks

Group Discussions/Technical Report Writing: 05 Marks.



Regularity and active participation: 05 Marks.

## **2. Mid Semester Exam (30 Marks)**

Mid semester examination will be based on 40% to 50% syllabus.

## **END SEMESTER EXAMINATION (50 MARKS)**

End Semester Examination will be based on syllabus coverage up to the Mid Semester Examination (MSE) carrying 20%-30% weightage, and the syllabus covered from MSE to ESE carrying 70%-80% weightage.